

ABSTRACT

HABITUATION-AWARE SIGNAL TEMPORAL LOGIC FOR PREDICTIVE MULTI-ROBOT BIRD DETERRENCE IN VINEYARDS

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Mobile multi-robot teams are a promising alternative to static scare devices for vineyard bird deterrence because they can continuously vary cue type and location, reducing habituation. A decentralized simulation testbed exists for this problem, but extensive sweeps show that its predictive dispatch branch fails to outperform a simple greedy baseline on the primary mission metric. The root cause is that the existing value function, a scalar exposure decrement ΔJ , captures only the immediate reduction from a single action, which greedy execution already optimizes. Because habituation makes action value depend on the full history of deployed cues and mode variety, ΔJ provides no planning signal for a history-aware planner to exploit. This thesis reformulates task value as the counterfactual robustness improvement of a Signal Temporal Logic (STL) specification $\Phi = \varphi_{\text{exp}} \wedge \varphi_{\text{cov}} \wedge \varphi_{\text{hab}}$, encoding exposure reduction, spatial coverage recency, and anti-habituation objectives via smooth conjunctive robustness. A per-cell habituation state, updated with multiplicative decrement κ and exponential recovery at time constant T_{rec} , is shared between the truth process and the planner’s forward predictor, and the resulting value slots into the existing reserved-capacity dispatcher without architectural change. On a six-baseline ladder (B0–B5) over ten seeds, the full system (B4) reduces mean value-weighted exposure by 6.8% ($p = 0.004$) relative to the legacy predictive baseline (B1) when habituation is active, and by 9.6% relative to the STL planner without the habituation clause (B3). Without habituation, B4 incurs roughly a 3% dispatch overhead, a cost that the 7% habituation-on gain more than offsets. These results offer simulation proof of concept for STL-based task valuation with a habituation clause, though no field validation is claimed and the habituation dynamics are not fitted to measured bird behavior. The framework extends to any multi-robot mission in which repeated cue use degrades local effectiveness.

Keywords: Signal Temporal Logic, multi-robot systems, habituation modeling, bird deterrence, predictive task allocation, spatio-temporal point process, robustness optimization.